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SUMMARY

ML researcher with three years of hands-on work designing and pretraining Vision-Language-Action foundation models from scratch. I'm the sole architect of an internal VLA family that reaches 95.7% average on LIBERO across all four suites, which puts us in the same range as published Pi0 results. The training corpus was 15,000+ hours of multimodal robot data: majority in-house, supplemented with Open X-Embodiment, DROID, and other open-source datasets. Before settling on the production architecture, I worked through every adjacent paradigm: LLM-as-planner, imitation learning, RL, diffusion, flow matching, and Drifting Models (arXiv 2602.04770). I also pretrained the VLM backbone from scratch on a 75TB multimodal dataset across 100+ GPUs at over 85% utilisation. Open to a senior IC research role or a tech lead role, depending on the team.

HIGHLIGHTS

- VLA architecture: designed and pretrained from scratch. 95.7% average on LIBERO across all four suites, in the same range as published Pi0 results.
- Trained on 15,000+ hours of multimodal robot data: majority in-house teleoperation and demonstration data, plus Open X-Embodiment, DROID, and other open corpora.
- Pretrained the VLM backbone from scratch on a 75TB multimodal dataset. 100+ GPUs on AWS SageMaker HyperPod at over 85% utilisation.
- Three years working through LLM-as-planner systems, imitation learning baselines (RT-1, RTX, Octo, Diffusion Policy), RL, diffusion, flow matching, and Drifting Models before settling on the architecture that worked.
- Built a hardware benchmarking suite from scratch so we could compare our models honestly against Pi0 and GR00T on real robots, not just in sim.
- First inventor on European Patent EP 4 550 261 B1 (granted March 2026).

SELECTED PROJECTS

Internal VLA model family. *Oct 2024 to present (v1 to v3)*

A Vision-Language-Action architecture I designed from scratch with action-decoding heads on top of our own pretrained VLM backbone. Trained on 15,000+ hours of multimodal robot data: majority in-house teleoperation and demonstration data, supplemented with Open X-Embodiment, DROID, and other open-source corpora. I tested several action decoders along the way (diffusion, behaviour cloning, flow matching, and Drifting Models per arXiv 2602.04770 for one-step inference). Flow matching gave the most stable training and shipped in production. The model now averages 95.7% on LIBERO across Goal, Object, Spatial, and Long suites, which puts us in the same range as published Pi0 results. We closed the gap against Pi0, Pi0.5, and GR00T N1/N1.5/N1.6 across our internal v1.0 to v3.0 releases. Validated on hardware; not in customer-facing production yet.

Internal VLM backbone. *Oct 2024 to Mar 2025*

The VLM on which the VLA family is built. Pretrained from scratch on our own multimodal data. I owned the full pretraining pipeline: data curation, distributed training on AWS SageMaker HyperPod, evaluation, and downstream integration into the VLA. Runs on the same 100+ GPU setup as the VLA training.

Earlier work that got me here. *2023 to 2024*

- LLM-as-planner systems (Voxposer-style): high-level plans from LLMs, low-level controllers executing. Demoed internally.
- Imitation learning baselines: ran RT-1, RTX, Octo, 3D Diffuser Actor, and Diffusion Policy on our hardware before designing our own.
- RL framework: built the training side, and the data needs ended up inspiring the data platform above. Also directed a junior engineer's roughly 10-month project on VLMs as reward models.

EXPERIENCE

Team Lead, Foundation Models for Robotics, Agile Robots SE, Munich. *March 2026 to present*

- Lead a 15-person team across research, training infrastructure, and evaluation. Hired most of them myself.
- Own the technical strategy and quarterly roadmap for the VLA, VLM, and Embodied Foundation Model programmes.
- Led training of the latest internal VLA release. We closed the gap significantly against Pi0, Pi0.5, and GR00T N1/N1.5/N1.6 across our v1.0 to v3.0 releases.
- Built a hardware benchmarking suite from scratch. We needed something honest to compare our models against Pi0 and GR00T on real robots, not just in sim.

Tech Lead, Foundation Models for Robotics, Agile Robots SE, Munich. *August 2025 to March 2026*

- Designed a Vision-Language-Action architecture from scratch with action-decoding heads. Pretrained it from scratch on internal multimodal data. Reached 95.7% average on the LIBERO benchmark across all four suites.

- Trained the VLA family on 15,000+ hours of multimodal robot data: majority in-house teleoperation and demonstration data, supplemented with Open X-Embodiment, DROID, and other open-source corpora.
- Tested diffusion, behaviour cloning, flow matching, and Drifting Models (arXiv 2602.04770) as action decoders. Flow matching gave the most stable training and shipped in production.
- Built an Embodied Foundation Model: a unified perception, reasoning, and action model similar in spirit to Generalist AI's approach. Evaluated on LIBERO.
- Pretrained the VLM backbone that the VLA family is built on, from scratch, on internal multimodal data.
- Ran distributed pretraining on AWS SageMaker HyperPod across 100+ GPUs at over 85% utilisation.
- Managed our 75TB multimodal dataset. Loading latency dropped about 60% after I rewrote the caching and S3 access patterns.

Senior Deep Learning Engineer / Acting Tech Lead, Agile Robots SE, Munich. *December 2023 to August 2025*

- Built a Transformer-based imitation learning policy. Over 85% success on unseen manipulation tasks from 5 to 20 demonstrations.
- Tested RT-1, RTX, Octo, 3D Diffuser Actor, and Diffusion Policy on our hardware. The lessons fed directly into our own VLA design choices.
- Built an RL training framework. Directed a junior engineer's roughly 10-month research project on VLMs as reward models for RL policy training.
- Validated LLM-as-planner systems (Voxposer-style): high-level plans from LLMs, low-level controllers executing.
- Built a simulation validation framework with Gym, ZeroMQ, and ReRun so we could check training data quality before kicking off pretraining runs.

Deep Learning Engineer, Agile Robots SE, Munich. *August 2021 to December 2023*

- Built a semi-supervised training framework that cut labelling overhead by about 60% while staying within 2% of the supervised baseline.
- Built a PyTorch detection library that ran at 45 FPS on edge hardware via structured pruning, INT8 quantisation, and TensorRT.
- Deployed TensorRT and ONNX models on robot hardware, achieving 3x to 5x inference speedups.

Master's Thesis, Multi-Object Tracking, Robert Bosch GmbH, Hildesheim. *September 2020 to March 2021*

- Extended the SHAMANN paradigm for tracking stability in crowded scenes. Matched SOTA on MOT17 and MOT20 with about 30% less model complexity.

Deep Learning Intern, Auvisus GmbH, Karlsruhe. *March 2020 to August 2020*

- Optimised mobile PyTorch models with ONNX Runtime and TensorRT 4x to 7x speedups at over 92% accuracy.
- Built MobileNetV2 classifiers for sparse real-world data (under 500 samples).

SKILLS

Robotics ML: Vision-Language-Action models, vision-language models, imitation learning, flow matching, diffusion policy. End-to-end from architecture to robot hardware.

Training infrastructure: Distributed pretraining on 100+ GPUs (AWS SageMaker HyperPod), mixed precision, data pipeline design at the 75TB and 15,000-hour scale.

Deployment: TensorRT, ONNX, INT8/FP16 quantisation, structured pruning. 3x to 7x inference speedups in production.

Frameworks: PyTorch (expert), JAX, TensorFlow, Hugging Face, LangChain, Gym, ReRun, OpenCV.

MLOps: Docker, Kubernetes, GitLab CI/CD, MLflow, AWS (S3, EC2, SageMaker), Ray Serve, ZeroMQ.

Leadership: Built and hired a 15-person team. Comfortable owning roadmaps, running hiring panels, and mentoring at all levels.

Programming languages: Python (expert), C++, CUDA tooling, Rust, Bash, SQL.

Spoken languages: English (fluent), Hindi and Telugu (native), German (basic).

EDUCATION

M.Sc. Computer Science (Digital Engineering), Otto-von-Guericke University Magdeburg. Grade 1.8.

B.Tech. Mechanical Engineering, JNTUA College of Engineering, Ananthapuram. 75%.

PATENT AND PUBLICATIONS

- European Patent EP 4 550 261 B1, "Animal Physical Parameter Estimation by Image Processing". Granted 18 March 2026 (application 23207432.8). First inventor; invented the ML and vision idea and drove the filing. Co-inventors: L. S. P. Kondamadugula, P. Topcam, K. Sharma, Z. Marton.
- Niemann A., Kandapagari P., et al. "Tissue Segmentation in Histologic Images of Intracranial Aneurysm Wall". Interdisciplinary Neurosurgery, Vol. 26 (April 2021).